5508 module04

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This program (created 2026-02-10 by Reagan Todd) shows a brief introduction to null values in SQL. This belongs in the public domain so please use however you see fit.

```
R.version.string
```

```
[1] "R version 4.5.2 (2025-10-31)"
```

```
Sys.Date()
```

```
[1] "2026-02-13"
```

```
library(tidyverse)
library(DBI)
library(RSQLite)
library(readr)
```

creating the MIMIC III database from the demo folder

create a new SQLite database file for patients table

```
con <- dbConnect(SQLite(), "mimic_demo.db")

patients <- read_csv(
  "~/Documents/Spring 26/MEDB-5530/data/mimic-iii-demo/PATIENTS.csv"
)
```

```
Rows: 100 Columns: 8
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr  (1): gender
dbl  (3): row_id, subject_id, expire_flag
dttm (4): dob, dod, dod_hosp, dod_ssn

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
dbWriteTable(con, "patients", patients, overwrite = TRUE)
```

Q1. Verify that gender in patients table has no missing values

```
cox <- dbConnect(SQLite(), "mimic_demo.db")
patients <- dbGetQuery(conn=cox, "
    select *
    from patients
    where gender is null;
")
patients
```

```
[1] row_id      subject_id  gender      dob         dod         dod_hosp
[7] dod_ssn     expire_flag
<0 rows> (or 0-length row.names)
```

```
dbDisconnect(cox)
```

The output of 0 rows confirms that there are no missing values for gender in the patient table.

Q2. Count the number of missing values for dod_ssn in patients table

```
cox <- dbConnect(SQLite(), "mimic_demo.db")
patients <- dbGetQuery(conn=cox, "
    select *
    from patients
    where dod_ssn is null;
")
patients
```

```
   row_id subject_id gender         dob        dod   dod_hosp dod_ssn
1    9472      10011      F  3800304000 4943548800 4943548800      NA
2    9522      10064      M  2786745600 4961088000 4961088000      NA
3    9525      10067      M  4147804800 5073148800 5073148800      NA
4    9553      10098      F  5709657600 6340464000 6340464000      NA
5    9556      10101      M  2699481600 4993142400 4993142400      NA
6    9572      10117      F  3229632000 5329324800 5329324800      NA
7    9575      10120      M  3734553600 4587321600 4587321600      NA
8   30839      40177      M  3549744000 6291216000 6291216000      NA
9   30876      40310      M  4226256000 5522428800 5522428800      NA
10  30924      40503      F  4035225600 6832512000 6832512000      NA
11  30962      40655      F -3959020800 5528131200 5528131200      NA
12  31314      41983      F -3733344000 5734022400 5734022400      NA
13  31323      42033      F  2390947200 5098636800 5098636800      NA
```

```
14  31332      42066      M  2885846400 4484592000 4484592000      NA
15  31372      42199      F  2350598400 4646937600 4646937600      NA
16  31391      42275      M  2795644800 5245776000 5245776000      NA
17  31427      42412      M  3442176000 6198076800 6198076800      NA
18  31440      42458      M -3895689600 5607187200 5607187200      NA
19  31752      43735      M  2414361600 5013014400 5013014400      NA
20  31758      43746      F  1891296000 4450636800 4450636800      NA
21  31789      43879      F  2573856000 4339958400 4339958400      NA
22  31870      44222      M  4338576000 6708614400 6708614400      NA
23  31872      44228      F  4506537600 6351868800 6351868800      NA
   expire_flag
1            1
2            1
3            1
4            1
5            1
6            1
7            1
8            1
9            1
10           1
11           1
12           1
13           1
14           1
15           1
16           1
17           1
18           1
19           1
20           1
21           1
22           1
23           1
```

```
dbDisconnect(cox)
```

There are 23 rows for dod_ssn where there is a null value. This means there are 23 missing or null values for dod_ssn in the patients table.

To answer Q3, we need to load in the DRGCODES.csv into the database.

```
con <- dbConnect(SQLite(), "mimic_demo.db")

drug_codes <- read_csv(
```

```
  "~/Documents/Spring 26/MEDB-5530/data/mimic-iii-demo/DRGCODES.csv"
)
```

```
Rows: 297 Columns: 8
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (3): drg_type, drg_code, description
dbl (5): row_id, subject_id, hadm_id, drg_severity, drg_mortality

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
dbWriteTable(con, "drug_codes", drug_codes, overwrite = TRUE)
```

Q3. Find the drg_code of the drgcodes table where drg_severity is null.

```
cox <- dbConnect(SQLite(), "mimic_demo.db")
drug_codes <- dbGetQuery(conn=cox, "
    select
      drg_code,
      drg_severity
    from drug_codes
    where drg_severity is null;
")
drug_codes
```

```
    drg_code drg_severity
1        148           NA
2        518           NA
3        185           NA
4        001           NA
5        205           NA
6        024           NA
7        138           NA
8        005           NA
9        115           NA
10       416           NA
11       089           NA
12       109           NA
13       218           NA
14       121           NA
15       010           NA
16       020           NA
17       320           NA
18       416           NA
19       079           NA
20       468           NA
```

21	416	NA
22	475	NA
23	527	NA
24	121	NA
25	416	NA
26	416	NA
27	541	NA
28	416	NA
29	403	NA
30	475	NA
31	416	NA
32	001	NA
33	174	NA
34	195	NA
35	486	NA
36	014	NA
37	468	NA
38	416	NA
39	300	NA
40	491	NA
41	105	NA
42	541	NA
43	300	NA
44	240	NA
45	578	NA
46	202	NA
47	202	NA
48	568	NA
49	578	NA
50	249	NA
51	210	NA
52	082	NA
53	123	NA
54	449	NA
55	014	NA
56	565	NA
57	487	NA
58	293	NA
59	003	NA
60	208	NA
61	175	NA
62	064	NA
63	870	NA
64	643	NA
65	388	NA
66	208	NA
67	689	NA
68	291	NA
69	689	NA
70	368	NA
71	853	NA

72	356	NA
73	871	NA
74	207	NA
75	085	NA
76	820	NA
77	432	NA
78	004	NA
79	291	NA
80	189	NA
81	604	NA
82	871	NA
83	329	NA
84	981	NA
85	326	NA
86	248	NA
87	657	NA
88	470	NA
89	238	NA
90	207	NA
91	823	NA
92	871	NA
93	283	NA
94	187	NA
95	083	NA
96	444	NA
97	190	NA
98	326	NA
99	871	NA
100	191	NA
101	280	NA
102	552	NA
103	236	NA
104	167	NA
105	682	NA
106	441	NA
107	314	NA
108	308	NA
109	287	NA
110	208	NA
111	189	NA
112	249	NA
113	441	NA
114	207	NA
115	840	NA
116	064	NA
117	444	NA
118	004	NA
119	193	NA
120	871	NA
121	853	NA
122	698	NA

```
123       871            NA
124       871            NA
125       871            NA
126       871            NA
127       394            NA
128       871            NA
129       871            NA
```

```
dbDisconnect(cox)
```

There are 129 null or missing values for drug severity and listed above are the drug codes (drg_code) for those missing drug severities.